Natacha **Kuete Meli**

Curriculum vitae

Objective

Dedicated and results-driven research scientist. Seeking to contribute to the research in the field of quantum computing, computer vision and image processing.

Profile

Name: Natacha **Kuete Meli**
Gender: Female
Nationality: Cameroon
Place of birth: Galim, Cameroon
Country of residence: Germany, since October 2017
Personal homepage: https://nkuetemeli.github.io

Language skills

French
C2
Mother tongue

English
Good

German
C1

Degrees

2021–2024
PhD in computational life science
University of Luebeck · Germany
Thesis: Quantum Algorithms for Binary Problems with Applications to Image Processing.
Supervisor: Prof. Dr. Jan Lellmann.
Variational quantum circuits • QUBO/Ising models • Quantum annealing.

2018–2021
Master in computational life science
University of Luebeck · Germany
Thesis: Trainable Detection Methods for Industrial Materials Testing.
Supervisor: Prof. Dr. Jan Lellmann.
Object detection • image segmentation • deep learning • R-CNN • U-Net.

2012–2015
Bachelor in mathematics and computer science
University of Dschang · Cameroon
Thesis: Backend Programming of a real estate web application in Java EE.
Supervisor: Prof. Maurice Tchoupe Tchendji.
Java EE • HTML • PHP • UML.

Competencies

Mathematics
Linear algebra • analysis • numerics • optimization • signal processing • image processing • graph theory • approximation theory • relativity theory • test and estimation theory • stochastic processes.
Quantum computing
Modeling and discrete optimization • QUBO/Ising models • quantum annealing • quantum circuits • variational quantum circuits • hybrid quantum algorithms • approximate quantum algorithms.
Image processing
Intensity transformations • morphological operations • principal component analysis • image reconstruction • active contour models • statistical shape models • atlas-based segmentation • parametric image registration • non-parametric image registration.
Machine learning
Image classification • object detection • image segmentation.

Academic Experience

April 2024 – August 2027
Researcher/PostDoc
University of Siegen · Germany
Supervisor: Prof. Dr. Michael Moeller

April 2021 - March 2024
Researcher/PhD Candidate
University of Luebeck · Germany
Supervisor: Prof. Dr. Jan Lellmann.

Research Interest

I specialize in applications of quantum computing to computer vision. This involves developing quantum algorithms for a range of computer-vision related optimization problems, both combinatorial and non-combinatorial. Quantum computing offers a powerful solution to overcome challenges posed by non-convex energies, limited storage, and high computational demands, promising more accurate and cost-effective solutions for real-world computer vision problems.

Programming skills

- MATLAB
- PYTHON
- C++
- JAVA
- LaTeX

Quantum programming

D-WAVE LEAP · QISKIT

Teaching Activities / Scientific assistant

Responsible for creating exercise sheets and conducting student assessments to complement lectures in the following subjects:

University of Siegen · Germany

Introduction to Quantum Computer Programming	Winter term 2025 and 2026
Non-linear convex optimization	Summer term 2025
Seminar: Recent advances in machine learning	Summer term 2025
Deep learning	Winter terms 2025 and 2026

University of Luebeck · Germany

Quantum image processing	Winter terms 2021, 2022 and 2023
Optimization	Summer terms 2022 and 2023
Linear algebra	Summer term 2021 and winter term 2023
Seminar: Numerical optimization	Winter terms 2021 and 2022
Mathematical methods of image processing	Winter term 2021
Computer graphics	Summer term 2020

University of Luebeck · Germany

Responsible for database maintenance, organization, and follow up of Erasmus applications as a scientific assistant at the International Office of the University of Luebeck.

Winter term 2019 - summer term 2021

Reviewing Activities

Participated in the following events and conferences as a reviewer:

ICML'26	International Conference on Machine Learning
ICLR'26	International Conference on Learning Representations
BMVC'25	British Machine Vision Conference
TPAMI'25	Transactions on Pattern Analysis and Machine Intelligence
3DV'25,'26	International Conference on 3D Vision
GCPR'24	German Conference on Pattern Recognition
QDSM'23, '24	International workshop on Quantum Data Science and Management
SSVM'23	International conference on Scale Space and Variational Methods in computer vision
CVPR'23,'25	IEEE/CVF conference on Computer Vision and Pattern Recognition
BVM'21	Bilbverarbeitung für die Medizin

Certificates

Continuously advancing my knowledge through online courses, I have earned the following certificates:

Sep. 2023
Qiskit global summer school 2023 - Quantum excellence
IBM · Online
For exploring the world of quantum computing through physics, mathematics, and python via Qiskit, to bridge the gap between quantum theory and real-world implementation.
Quantum teleportation • iterative phase estimation • qiskit runtime.

Aug. 2023
QBronze
QWORLD · Online
For successfully completing the online training in Quantum Computing & Programming using QWorld's introductory tutorial Bronze-Qiskit in the Womanium Global Quantum Program.
Quantum entanglement • quantum search • quantum annealing.

Mai 2023
IBM Quantum Challenge: Spring 2023
IBM · Online
For understanding how to create circuits that perform mid-circuit measurements and dynamically decide what the next steps should be.
Dynamic circuits • phase estimation • quantum teleportation • error correction.

Research skills

Oral presentation

Literature review

Academic writing

Team work

Connect

Google scholar • ResearchGate • ORCID • GitHub • Personal homepage

Conferences

Presented research findings at the following conferences:

3DV 2026
International Conference on 3D Vision
Vancouver · Canada

ECCV 2024
European Conference on Computer Vision
Milan · Italy
Talk: Quantum Binary Optimization.

MIA 2023
Mathematics and Image Analysis
Berlin · Germany
Poster: Iterative quantum transformation estimation.

CVPR 2022
Computer vision and pattern recognition
New orleans · USA
Poster: An iterative quantum approach for transformation estimation from point sets.

Students

Supervised final thesis of the following students:

April. 2025
Prajwal Maruti Edhate (M.Sc.)
University of Siegen · Germany
Thesis: Large-Scale 3D Point Cloud Classification using Parameterized Quantum Circuits.
Parameterized quantum circuits • point cloud classification • data re-uploading PQCs.

Nov. 2023
Josephine Elisabeth Oettinger (B.Sc.)
University of Luebeck · Germany
Thesis: Non-Boolean quantum amplitude amplification for discrete optimization.
Quantum search • Ising model • discrete optimization.

Volunteer

Worked on a voluntary basis on the following projects:

2022-2026
Mentoring-programm CyberMentor Plus
Germany · Online
Promoted science and provided online mentoring to 8th-11th-grade schoolgirls. Served as a STEM model (Mentor) and supervised the following STEM projects:

- Understanding online shops (2023)
- Fractals, Mandelbrot sets and visualization (2024)
- Digital image processing with Python (2023, 2024)
- Deep learning and for Image Classification (2025-2026)

Engagements

Have been an engaged member in the following programs:

2024–2027
Women in Vision Siegen
Germany · University of Siegen
Promoted women's academic careers in computer vision and related sciences at the University of Siegen, organizing discussions and activities involving women.

2026–2027
FraMeS: Frauenspezifisches Mentoring Siegen
Germany · University of Siegen
I have been a mentee in the FraMeS program under the mentorship of Dr. Maria Schuld (University of KwaZulu-Natal, South Africa). Our mentoring discussions covered both conceptual topics (quantum machine learning) and career-oriented aspects (women's academic career development).

Personal Interests

Reading • Sewing clothes • Running

Peer-reviewed publications

Authored or co-authored the following peer-reviewed publications:

- Johannes David Voigts · **Natacha Kuete Meli** · Jan Lellmann. Quantum Hamiltonian Descent for Rigid Image Registration. International Conference on Pattern Recognition (ICPR), 2026. #.
- **Natacha Kuete Meli** · Jovita Lukasik · Vladislav Golyanik · Michael Moeller. Layered Quantum Architecture Search for 3D Point Cloud Classification. International Conference on 3D Vision (3DV), 2026. https://arxiv.org/pdf/2603.20024.
- Shuteng Wang · **Natacha Kuete Meli** · Vladislav Golyanik · Michael Moeller. Quantum Multiple Rotation Averaging. International Conference on 3D Vision (3DV), 2026. https://arxiv.org/pdf/2602.10115.
- **Natacha Kuete Meli**, Vladislav Golyanik · Marcel Seelbach Benkner · Michael Moeller. QuCOOP: A Versatile Framework for Solving Composite and Binary-Parametrised Problems on Quantum Annealers. Conference on Computer Vision and Pattern Recognition (CVPR), 2025. https://doi.org/10.1109/CVPR52734.2025.01064.
- **Natacha Kuete Meli** · Florian Mannel · Jan Lellmann. A universal quantum algorithm for weighted maximum cut and Ising problems. Quantum Inf Process, 2023. https://doi.org/10.1007/s11128-023-04025-x.
- **Natacha Kuete Meli** · Florian Mannel · Jan Lellmann. An iterative quantum approach for transformation estimation from point sets. Conference on Computer Vision and Pattern Recognition (CVPR), 2022. https://doi.org/10.1109/CVPR52688.2022.00061.

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